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Delayed retirement and the social wealth distribution in China: an evaluation of policy effects

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China's delayed-retirement reform—recently announced and scheduled to start in 2025—seeks to bolster public-pension sustainability, yet its implications for wealth distribution remain unclear. Using six waves of the China Family Panel Studies (CFPS 2010–2020), we estimated a micro-level wealth-accumulation equation and embedded it in an agent-based counterfactual simulation. We examined three policy scenarios: a gradual schedule calibrated to the officially announced plan but re-indexed to begin in 2021 for the counterfactual window, and two comparators—a no-delay baseline and a one-off delay that immediately meets the target statutory ages. Using the 2020 CFPS wave (the latest available) as the baseline micro-dataset, we conduct an ex-ante counterfactual experiment that applies the reform as if it started in 2021; under a short-run, partial-equilibrium, full-compliance closure over a common 2021–2025 window (under 2020 conditions), the simulation traces changes in the national wealth distribution and the structural channels through which they propagate. Within our ex-ante counterfactual setting (2021–2025 under 2020 baseline conditions) and a short-run, partial-equilibrium, full-compliance closure, delayed retirement is associated with a higher wealth-inequality trajectory over 2021–2025, chiefly via widening gaps linked to urban–rural residence and health status; gender-related contributions remain comparatively small. Urban residents and healthier individuals are more likely to convert longer careers into additional wealth and pension entitlements, whereas gains among rural and less-healthy groups appear more limited. The findings are consistent with social-stratification and cumulative-advantage perspectives in China and present a transparent, ex-ante evaluation framework that can help inform the design and calibration of complementary measures in forthcoming pension reforms. These results should be interpreted as counterfactual policy-path differences over a common five-year horizon, not as realised outcomes or long-run forecasts.

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Introduction

The accelerated pace of global population ageing has posed substantial challenges for many countries, including increased burdens on pension systems, a reduced labour supply and slower economic growth. Delayed retirement policies, intended to extend labour force participation, aim to alleviate pension pressures, augment the labour supply and maintain economic vitality. As such, these policies have become a crucial strategy in addressing the challenges of an ageing population. In recent years, several member states of the Organisation for Economic Co–operation and Development (OECD), including the United States, Germany and Japan, have implemented delayed retirement measures. Similarly, international organisations such as the World Bank and the International Labour Organization have advocated for these policies to mitigate socioeconomic pressures (Clemens and Parvani 2017).

In China, the statutory retirement ages are currently 60 years for men, 55 years for female cadres (white-collar) and 50 years for female workers (blue-collar). A Standing Committee resolution (September 2024) mandates a month–by–month increase from 2025, so that the thresholds will reach 63, 58 and 55 years, respectively, over the next fifteen years (MoHRSS (China) 2024). Chinese basic pension coverage already exceeds 95% of the pension–age population, yet benefits remain highly unequal across income groups, gender and regions (Yicai 2025). A gradual age extension is projected to lengthen contribution spells and raise the average replacement rate for formal–sector workers from 42% to about 48% by 2040 (People's Daily Overseas Edition 2021). Because the latest nationally representative microdata available to us are from 2020, our evaluation is ex ante: although the statutory reform is scheduled to start in 2025, we re-index (shift) the official month-by-month age-increase schedule to begin in 2021 for a counterfactual experiment, holding 2020 socioeconomic conditions fixed (see Section 4.3.2).

Wealth inequality has become one of the most significant forms of inequality in twenty–first century (Piketty 2014). It not only triggers financial crises and suppresses economic growth but also exacerbates social tensions, undermining both economic and social stability (Aaskoven 2022; Bagchi and Svejnar 2015). In China, the issue of income and wealth inequality has become increasingly prominent. Since the reform and opening up, rapid economic growth has driven the accumulation of wealth at both the family and individual levels. The annual growth rate of per capita net assets has far surpassed that of residents' income during the same period (Li et al. 2023). The Gini coefficient for social wealth has remained persistently high, further deepening wealth inequality (Liu and He 2024; Wan et al. 2021).

Previous studies have indicated that delayed retirement may significantly impact wealth accumulation through its effects on individual labour supply, income and savings behaviour (Börsch-Supan and Ludwig 2010). However, these effects are not uniformly distributed across different social groups, as factors such as health status, income levels and educational background may lead to heterogeneous policy outcomes (Rogerson and Wallenius 2009; Shoven and Slavov 2014). Nevertheless, analysing the determinants of income and wealth at the micro level does not directly allow us to infer changes in wealth distribution at the macro level. This micro–to–macro analytical framework is essential for a comprehensive understanding of the societal impacts of delayed retirement policies. Therefore, this paper examines how delayed retirement policies affect social wealth distribution at the macro level through individual differences, providing a more holistic perspective for policy formulation. This approach not only deepens our understanding of delayed retirement policies but also offers an ex-ante, model-based assessment of potential distributional consequences in China, which may inform discussions of complementary measures aimed at improving equity in wealth outcomes.

Literature Review

Causes and consequences of wealth inequality. In recent years, wealth inequality has evolved into a central issue affecting both economic and social development, attracting scholarly attention. With the acceleration of globalisation and shifts in social structures, wealth inequality has become increasingly pronounced, posing risks of social discontent and economic instability in many countries (Atkinson 2015; Piketty 2014). Consequently, researchers have systematically explored the causes and consequences of wealth inequality.

Early research primarily focused on income inequality. However, as scholars undertook in–depth analyses of long–term wealth trends, they shifted their focus to wealth inequality as a distinct issue. Wealth inequality is not merely an economic phenomenon; it is also closely intertwined with social structures, policy directions, historical contexts and cultural factors. Through a comprehensive analysis of wealth inequality trends since the eighteenth century, Piketty (2014) revealed the historical roots of wealth concentration and its relationship with the rate of return on capital, highlighting the critical role of policy in shaping wealth distribution. Furthermore, the exacerbation of wealth inequality may lead to social unrest and pose a threat to democratic institutions (Aaskoven 2022), further emphasising the broader societal significance of this issue.

Empirical studies have examined multiple factors influencing wealth inequality, such as disparities in educational opportunities, changes in market structures, policy orientations and the global economic environment (Saez and Zucman 2016; Wolff 2017). These factors interact in complex ways, leading to unequal wealth accumulation and distribution, which, in turn, affect overall social stability and economic development (Piketty et al. 2019). It is evident that the formation and evolution of wealth inequality are a complex process, requiring a comprehensive consideration of economic, social and policy factors to deepen our understanding of wealth distribution mechanisms.

Current research on wealth inequality in China. In recent years, wealth inequality in China has not only persisted but has also been exacerbated by the development of the market economy (Zhao et al. 2024). Since the initiation of economic reforms, rapid capital accumulation and a sharp rise in real–estate prices have become key drivers of the growing wealth disparity (Wan et al. 2021). The marketisation process has led to the swift accumulation of wealth among the top 10% of households, while the wealth share of the bottom 50% has substantially decreased, creating a highly unequal distribution pattern (Piketty et al. 2019). According to statistics, the Gini coefficient for household net assets in China reached approximately 68% in 2020 (Liu and He 2024). This trend has drawn significant attention from both scholars and policymakers, as the high concentration of wealth not only hinders economic growth but also poses a potential threat to social stability.

Rapid urbanisation has been successful in reducing absolute poverty but has, to some extent, exacerbated urban–rural wealth inequality. Due to policy biases and the concentration of resources, urban areas have absorbed a disproportionate share of wealth, while rural labourers and low–income families have struggled to fully benefit (Chen et al. 2011; Liu et al. 2017). The centralised allocation of urban infrastructure, education and healthcare resources has placed rural areas at a disadvantage in terms of social security and public services

(Li and Jin 2022), further limiting the access of rural households to educational and development opportunities, thereby deepening wealth–distribution disparities.

The boom in the real–estate market is one of the primary drivers behind rapid urban–wealth accumulation; urban households derive the majority of their wealth from property appreciation (Yue and Ying 2022). Data indicate that a substantial portion of urban household assets is concentrated in real estate, whereas rural areas, due to insufficient capital accumulation, are unable to partake in this appreciation process (Yao et al. 2024). This disparity has significantly widened the wealth gap between urban and rural areas, resulting in a polarised wealth distribution.

Despite the implementation of a series of policies in recent years, such as tax reforms and increased social–welfare spending, which have somewhat alleviated wealth inequality (Wu and Zhang 2024), overall wealth distribution remains relatively imbalanced. The processes of marketisation and urbanisation have further concentrated wealth among a small group, while the relatively low emphasis on personal income tax has limited the capacity for wealth redistribution (Jain-Chandra et al. 2018). Inequality in educational opportunities has further exacerbated this phenomenon, as individuals with higher levels of education are more likely to secure high–income occupations, thereby intensifying the intergenerational transmission and accumulation of wealth inequality (Wu 2017). Consequently, to effectively mitigate wealth inequality, comprehensive reforms in policy, education and the tax system are urgently required.

Socioeconomic impacts of delayed retirement. As a key policy measure to address population ageing, the socio–economic impacts of delayed retirement have attracted considerable attention. Research indicates that delayed retirement has profound effects on multiple areas, including labour markets, pension systems and the distribution of social wealth.

Firstly, extending the retirement age can increase the labour supply, alleviating the labour shortages induced by population ageing and, consequently, promoting economic growth (Gruber and Wise 2010). Furthermore, an extended working life may incentivise workers to improve their skills and invest in human capital, thereby enhancing overall labour productivity. However, this could also lead to increased competition between older and younger workers in the labour market, potentially reducing youth employment rates and career development opportunities (Boeri et al. 2016).

Secondly, regarding pension burdens, delayed retirement helps ease the financial pressures on pension systems. By postponing the age at which pensions are collected, the duration of pension payments is shortened while the contribution period is lengthened, which enhances the sustainability of pension systems and reduces the fiscal burden on governments (Börsch-Supan and Coile 2018). However, for workers engaged in physically demanding jobs or those with poor health, delayed retirement may increase physical strain, negatively affecting health and quality of life. As such, pension policy design should take into account the varying needs of different groups to ensure the fairness of social–welfare provisions (OECD 2023).

Thirdly, delayed retirement is poised to reshape wealth distribution through two intertwined channels. Within the pension system, China's fragmented, earnings–related schemes disproportionately bolster the entitlements of urban formal–sector men. Empirical evidence indicates that public pension income is more unequally distributed than labour earnings (Li et al. 2020); women receive barely half the benefit of men (Zhao and Zhao 2018) and rural or informal workers accumulate minimal pension wealth (Wu 2013). Extending the statutory age without reforming these rules therefore risks amplifying existing disparities.

Moreover, delayed retirement could influence the overall pattern of wealth distribution, concentrating wealth further among high–income and healthier groups, thereby potentially increasing wealth inequality (Rasmussen 2018; Xie et al. 2024). It may also affect intergenerational wealth distribution by delaying job vacancies, which impacts employment and promotion opportunities for younger generations, ultimately influencing their income and wealth accumulation (Bertoni and Brunello 2017). This, in turn, could lead to an unequal distribution of wealth across generations, affecting intergenerational equity within society (Piketty 2014).

In conclusion, while existing research has deepened our understanding of wealth inequality, evidence on how delayed retirement translates into distributional change remains limited in China, where the reform is nascent. Delayed retirement can reshape both wealth accumulation and its cross–sectional distribution through three proximal channels: (i) longer labour earnings and contribution spells with a later conversion of wages into pension income; (ii) earnings-related, scheme-differentiated replacement rules that produce heterogeneous gains across urban/formal versus rural/informal groups and by gender and health; and (iii) short-run labour-market crowding and household portfolio/liquidity responses. Building on these mechanisms, we focus on the short run (2021–2025) and examine how policy-induced shifts in the timing of the wage-to-pension transition map into the cross-sectional distribution of social wealth across individuals.

Theoretical Analysis

To investigate the impact of delayed retirement policies on the distribution of social wealth, a dynamic stochastic general equilibrium (DSGE) model is constructed. This model incorporates a delayed–retirement shock to examine its effect on individual wealth accumulation and to derive the evolutionary path of overall social–wealth distribution.

Assume the economy consists of two types of individuals: high–asset and low–asset groups. Each group has a lifespan of T periods, faces the same interest rate r and shares a constant risk–aversion coefficient σ.

The utility function for an individual is given by Hall (1988):

$$U^i = \sum_{t=0}^{T} \beta^t \left[\frac{\left(C_t^i\right)^{1-\sigma}}{1-\sigma} - \psi \frac{\left(L_t\right)^{1+\eta}}{1+\eta} \right]$$

where $\beta \in (0,1)$ is the time preference factor, C_t^i denotes consumption in period t and L_t is the labour supply in period t (which is assumed identical across individuals). The parameters $\psi > 0$ and $\eta > 0$ denote parameters in the disutility–of–labour function.

The budget constraint for an individual is given by Blanchard and Fischer (1989):

$$A_{t+1}^i = (1+r)A_t^i + wL_t - C_t^i$$

where A_t^i represents the wealth stock in period t, C_t^i is consumption in period t, L_t is the labour supply, w is the wage rate and r is the interest rate.

An individual maximises utility by choosing optimal consumption(C_t^i) and labour supply(L_t). Based on optimisation conditions, the Euler equation for consumption is derived as Hall (1978):

$$\left(C_t^i\right)^{-\sigma} = \beta(1+r)\left(C_{t+1}^i\right)^{-\sigma}$$

The optimal labour supply condition can be derived from the utility maximisation problem. The first–order condition is:

$$\frac{\partial U^i}{\partial L_t} = 0 \Rightarrow w = \psi(L_t)^{\eta}$$

This condition defines the optimal labour supply, indicating that the wage rate is related to the labour supply according to this relationship.

By substituting the optimal labour supply condition($w = \psi(L_t)^{\eta}$) into the budget constraint, the wealth accumulation equation is obtained:

$$A_{t+1}^i = (1+r)A_t^i + \psi(L_t)^{\eta}L_t - C_t^i$$

Here, L_t represents the optimal labour supply, which can be expressed by the optimal condition as:

$$L_t = \left(\frac{w}{\psi}\right)^{1/\eta}$$

Thus, the wealth accumulation equation becomes:

$$A_{t+1}^i = (1+r)A_t^i + \psi\left(\left(\frac{w}{\psi}\right)^{1/\eta}\right)^{\eta} - C_t^i$$

Define the wealth difference between high–asset and low–asset groups as:

$$\Delta A_t = A_t^H - A_t^L$$

From the wealth accumulation equation, the dynamic evolution of wealth differences is given by:

$$\Delta A_{t+1} = (1+r)\Delta A_t - \Delta C_t$$

where $\Delta C_t = C_t^H - C_t^L$ represents the consumption difference.

Using the Euler equation for consumption, the consumption difference can be expressed as:

$$\Delta C_t = C_0\left[\left(\beta^H(1+r)\right)^{t/\sigma} - \left(\beta^L(1+r)\right)^{t/\sigma}\right]$$

Substituting this into the wealth difference equation yields:

$$\Delta A_{t+1} = f(\beta, r, \sigma)\cdot \Delta A_t$$

where:

$$f(\beta, r, \sigma) = \frac{(1+r)^{1+1/\sigma}\beta^{1/\sigma}}{1+(1+r)^{1/\sigma}\beta^{1/\sigma}}$$

This equation suggests that the growth rate of wealth disparities is influenced by the time preference parameter β, the interest rate r and the relative risk aversion coefficient σ.

Given the constancy of the interest rate and the relative stability of the risk aversion coefficient, the time preference parameter β becomes a critical factor potentially subject to influence by delayed retirement policies. A higher β indicates a greater emphasis on future consumption, leading to a greater tendency for increased savings. Delayed retirement policies, by extending the working years, may alter individuals' consumption and saving behaviors, thereby affecting their time preferences (Hurd and Rohwedder 2012).

From the wealth difference equation, it is evident that even if delayed retirement policies increase the time preference parameter β by the same amount for both high–asset and low–asset groups, the wealth gap may still widen. This is because the growth factor $f(\beta, r, \sigma)$ in the wealth difference equation increases with β and the wealth gap is also influenced by the wealth base effect ΔA_t. Therefore, any direct or indirect impact of delayed retirement policies on the time preference parameter β (Moore et al. 2011) may alter the dynamic trajectory of social wealth distribution.

Empirical Research Design

To validate the internal logic underlying the theoretical analysis presented above, this study draws on data from the China Family Panel Studies (CFPS) and applies a dynamic panel model to capture the dynamic characteristics of individual wealth changes. In addition, based on microeconomic theory, a calibration approach using a structural model is employed to evaluate the potential impact of delayed retirement policies on individual wealth, thereby inferring their policy effects on the distribution of social wealth. Given that delayed retirement policies have been recently introduced, actual data cannot yet directly reflect their implementation effects. Therefore, counterfactual simulations are conducted to predict potential changes that the policy may bring, compensating for the limitations of real–world data.

Relative to alternatives in the public-pension literature, our design is tailored to a short, common horizon (2021–2025) and current data constraints. Administrative microsimulation would require individual contribution histories and accrual records that are not available in the CFPS; life-cycle or OLG general-equilibrium models demand strong, hard-to-verify assumptions over a brief roll-out period; and quasi-experimental designs are infeasible at this early stage of implementation. We therefore combine an estimated wealth-accumulation equation with a Mesa agent-based simulation and a mixed-kernel mapping from pre-retirement wages and occupations to expected first-period pension receipts. This short-run, partial-equilibrium closure preserves like-for-like comparability across scenarios, as detailed below.

Data description. The CFPS was launched in 2010 and conducted five follow–up surveys in 2012, 2014, 2016, 2018 and 2020. This survey targets a sample of 16,000 households from over 160 districts across 25 provinces (including municipalities and autonomous regions) in China, providing detailed records on key economic information such as individual and household income, consumption and assets (Xie and Hu 2014). The six–wave CFPS dataset utilized in this study represents the largest, most comprehensive and longest–running micro–level survey on wealth distribution in China to date.

Before conducting the empirical analysis, the data underwent the following preprocessing steps:

i. Sample selection: Households with a head of household under the age of 16 were excluded to ensure that the subjects had the capacity for labour and investment.
ii. Data deflation: Economic variables were deflated using the Consumer Price Index (CPI) for each province and year, with 2010 as the base year, to eliminate the impact of inflation.
iii. Sample weighting: The raw samples were weighted using the sampling weights provided by the CFPS database to ensure national representativeness and cross–year comparability.
iv. Outlier treatment: A 1% winsorisation (replacing extreme values with the nearest less extreme values) was applied to variables such as individual net assets, personal income and personal consumption to minimize estimation biases caused by extreme values.
v. Missing value handling: Samples with missing values were excluded to build a balanced 2010–2020 panel and ensure consistent lag structures. Baseline balance checks (Supplementary Table S1) show all standardized mean differences < 0.07 and IPW–IV estimates (Table 4) remain within 10% of the baseline, indicating that complete–case analysis does not materially bias our results.

After these preprocessing steps, approximately 10,906 tracking samples per wave were retained, resulting in a detailed and representative six–period panel dataset.

Variable Definition and Measurement. To accurately assess the impact of delayed retirement policies on wealth distribution, the following key variables were selected and measured, while accounting for individual heterogeneity. We use "social wealth (distribution)" to denote the cross-sectional distribution of adult-equivalised household net assets across individuals, which are measured as real and financial assets net of liabilities, excluding public capital and unobserved pension entitlements.

Wealth

Following Wan et al. (2021), we measure individual wealth as household net assets divided by the household's *equivalent–adult size* ($\text{EA_Size}_i(t)$). Formally,

$$\text{Wealth}_i(t) = \frac{\text{TotalAssets}_i(t) - \text{TotalLiabilities}_i(t)}{\text{EA_Size}_i(t)}$$

where *TotalAssets* comprise liquid deposits, securities and real estate, while *TotalLiabilities* include all outstanding loans and credit–card debt. $\text{EA_Size}_i(t)$ is constructed using the OECD–modified equivalence scale (OECD 2013), using the CFPS age cut–off:

$$\text{EA_Size}_i(t) = 1.0 + 0.5(Adults_{it} - 1) + 0.3 \times Children_{it}$$

assigning a weight of 1.0 to the first adult (≥ 16 years), 0.5 to each additional adult and 0.3 to each child (< 16 years).

Income

Total personal income consists of four components reported in the CFPS 'total personal income' file: wage income (earnings from paid employment), transfer income (government subsidies, social assistance and private remittances), business income (net profits from farming, self–employment and private enterprises) and asset income (rents, proceeds from property sales or leases, interest on deposits and returns on financial investments). For respondents who are retired, pension income is additionally included. Formally, for individual i in year t,

$$\text{Income}_i(t) = \text{WageIncome}_i(t) + \text{TransferIncome}_i(t) + \text{BusinessIncome}_i(t) + \text{AssetIncome}_i(t) + \text{Pension}_i(t) \cdot \mathbf{1}_{\{\text{Retired}_i(t)=1\}}$$

where $\mathbf{1}_{\{\text{Retired}_i(t)=1\}}$ is an indicator equal to 1 when the respondent is retired, so pension receipts are added only for retirees.

Consumption

Since complete individual consumption data are unavailable in the CFPS dataset, this study uses total household consumption divided by EA_Size as a proxy for individual consumption, calculated as follows:

$$\text{Consumption}_i(t) = \frac{\text{Total Household Consumption}_i(t)}{\text{EA_Size}_i(t)}$$

Although there may be heterogeneity in intra–household resource allocation, this indicator provides a reasonable approximation of individual consumption levels.

Control Variables

To account for other factors influencing wealth accumulation, the following control variables are included:

Urban: Type of residence (urban = 1; rural = 0).

Gender: Gender of the individual (male = 1; female = 0).

Age: Age of the individual.

Table 1 Descriptive Statistics of Variables.

Variable (unit)	Mean	Variance	Min	Max
Individual Net Assets	16.82	645.12	0.23	403.38
Total Personal Income	4.37	105.27	0.04	180.00
Wage Income	3.93	74.15	0	105.39
Pension Income	2.68	13.06	0	8.16
Personal Consumption	2.89	174.11	0.11	233.93
Household Net Assets	49.82	728.83	0.46	801.30
Household Income	5.43	109.05	0.14	233.60
Self–Assessed Health (1–5)	2.93	1.29	1	5
Urban–Rural (Urban=1)	0.45	0.49	0	1
Age (years)	50.09	14.39	16	100
Gender (male=1)	0.47	0.49	0	1
Marital Status (married = 1)	0.42	0.52	0	1

The analytical sample comprises 65,436 observations (10 906 individuals observed across six CFPS waves).
Monetary values are expressed in units of 10,000 yuan (CNY).
Urban–rural status, gender and marital status are binary variables coded 1 = yes, 0 = no.

Self_Rate_Health: Self–reported health status (very healthy = 1; healthy = 2; average = 3; unhealthy = 4; very unhealthy = 5).

Marital_Status: Marital status (married = 1; unmarried = 0).

All economic indicators were standardized to annual values to ensure cross–year comparability. Table 1 reports the basic statistics for the variables mentioned above.

Empirical model and method

Individual wealth accumulation model. **1. Model specification**. For each individual i at time t, we specify the following dynamic panel model of wealth accumulation:

$$Wealth_{i,t} = \alpha_0 + \alpha_1 Wealth_{i,t-1} + \alpha_2 Income_{i,t} + \alpha_3 Consumption_{i,t} + \gamma X_{i,t} + \delta_{r\times t} + \mu_i + \varepsilon_{i,t}$$

In this equation, $Wealth_{i,t}$ represents individual i's wealth at time t (the dependent variable)and $Wealth_{i,t-1}$ is the one–period lag of wealth capturing persistence. $Income_{i,t}$ and $Consumption_{i,t}$ denote individual i's income and consumption at time t, respectivelyand $X_{i,t}$ is a vector of additional control variables (with an associated coefficient vector γ). We include μ_i to capture unobserved individual–specific fixed effects and $\delta_{r\times t}$ to account for region–by–time fixed effects (common shocks specific to region r in period t). Finally, $\varepsilon_{i,t}$ represents the idiosyncratic error term.

Individual wealth tends to be highly persistent over time, making a dynamic panel specification appropriate. Omitting the lagged wealth term would cause model misspecification and omitted variable bias, because past wealth strongly influences current wealth (Wooldridge 2010). Including $Wealth_{i,t-1}$ as a regressor captures this intertemporal dependence, but its presence in a fixed–effects model introduces an additional challenge: the lagged dependent variable is correlated with the individual fixed effect, leading to bias in the coefficient estimates when the time dimension is short (Nickell 1981). This finite–sample bias—known as the Nickell bias—implies that standard fixed–effects estimators are inconsistent in dynamic panels. To address this issue, specialized dynamic panel estimators are employed to obtain consistent estimates in the presence of a lagged dependent variable.

2. Endogeneity. In an individual–level dynamic wealth model, income, consumption and past wealth are all endogenous variables. Income and consumption are choice variables

determined jointly with wealth accumulation, so they can be correlated with unobserved factors such as individual preferences or shocks (omitted variables) that also affect wealth. For example, an unobserved preference for saving could influence both consumption and wealth and simultaneity arises because current wealth depends on income minus consumption while those decisions might depend on expected wealth. Moreover, lagged wealth $Wealth_{i,t-1}$ is endogenous because it is correlated with the error term through unobserved individual effects and past shocks, a well–known dynamic panel bias (Nickell bias) in fixed–effects models. This means that ordinary least squares or even fixed–effects OLS estimates would be biased and inconsistent (Nickell 1981). Hence, endogeneity must be addressed to obtain consistent causal estimates.

A fixed–effects two–stage least squares (FE–2SLS) estimator is an appropriate solution to these endogeneity concerns. The fixed–effects framework controls for time–invariant individual heterogeneity (e.g. innate thrift or ability), eliminating bias from omitted stable traits. Two–stage least squares then uses external variation from instrumental variables (IVs) to isolate the independent changes in income, consumption and lagged wealth that are not driven by simultaneity or omitted variables (Judson and Owen 1997). By using FE–2SLS, we obtain consistent parameter estimates under standard IV assumptions (relevance and exogeneity), even in the presence of endogenous regressors (Angrist and Krueger 2001). This approach is analogous to established dynamic panel IV methods: for instance, Anderson and Hsiao (1981) propose a fixed–effects IV estimator to correct dynamic panel bias and Arellano and Bond (1991) develop a GMM procedure along similar lines. In our context, the FE–2SLS estimator leverages panel data by using variation within individuals over time, combined with valid instruments, to address reverse causality and measurement error. It thus effectively tackles both the endogeneity of regressors and unobserved heterogeneity in the wealth model, yielding more credible estimates of how income, consumption and past wealth influence current wealth.

3. IV Strategy. To mitigate endogeneity we employ three excluded instruments, each supported by recent empirical evidence.

First, *RetAge_Hit* is a binary indicator that equals one once an individual crosses China's mandatory–retirement threshold—60 for men, 55 for female cadres and 50 for female workers. Passing this legal cut–off produces an exogenous surge in the probability of labour–force exit, which in turn simultaneously depresses earned income and induces downward adjustments in consumption, yet leaves pre–existing net worth mechanically unchanged because China's scheme does not mandate lump–sum pension withdrawals or forced asset conversions at retirement. Hence we treat *RetAge_Hit* as an excluded instrument for both Income and consumption.

A growing body of research confirms the instrument's relevance and exogeneity: Fischer and Müller (2020) show that early–retirement thresholds shape labour supply and spending but not wealth accumulation; Olafsson and Pagel (2024) document discrete consumption drops at pension–eligibility ages driven by income shocks rather than direct wealth effects; and Deshpande et al. (2024) demonstrate that statutory eligibility alters cash–flow timing without affecting households' asset positions. These findings collectively validate our assumption that crossing the retirement–age threshold influences wealth only through its impact on income and consumption decisions.

Second, *PeerWage*—the leave–one–out mean wage of working–age individuals in the same community–year—is employed as an excluded instrument for both income and consumption. Localised firm– or sector–specific shocks shift neighbours' earnings, which (i) bolster the individual's wage–bargaining position (relevance for income) and (ii) alter reference norms and liquidity expectations (relevance for consumption). After saturating the specification with region × year fixed effects, a flexible age profile and cohort controls, no remaining channel links *PeerWage* to household wealth other than through contemporaneous income and consumption, meeting the exclusion restriction.

The design also tackles the classic reflection problem (Manski 1993). By excluding household from the community average and purging common shocks via high–dimensional fixed effects, simultaneity between the individual's choices and the peer instrument is minimised (Borusyak et al. 2022). Extensive evidence from social–interaction econometrics confirms that such leave–out group means are credible instruments under these conditions (Bramoullé et al. 2020).

Third, we include $Wealth_{i,t-2}$ – the individual's wealth lagged two periods – as an internal IV in a dynamic panel model. This follows the Arellano and Bond (1991) difference–GMM approach, which treats past levels of the outcome as instruments for current changes under the assumption that there is no second–order serial correlation in the errors. In other words, if the error term $\varepsilon_{i,t}$ has no autocorrelation beyond the first order, then $Wealth_{i,t-2}$ is uncorrelated with $\varepsilon_{i,t}$ and can serve as a valid instrument for $Wealth_{i,t-1}$ (Farzana et al. 2024). We verify this condition in our data: the Arellano–Bond test for AR(2) yields a p–value well above conventional significance levels, indicating no evidence of second–order error correlation. This result ($p > 0.5$) implies that the moment conditions hold and our use of the twice–lagged wealth is exogenous (Arellano and Bond 1991). The validity of such GMM instruments is also affirmed in recent research – for instance, Alvarez and Arellano (2022) develop a robust likelihood estimator for dynamic panels and emphasize checking AR(2) to ensure the integrity of internal instruments. In sum, the Arellano–Bond tests in our estimation are not significant at the second lag, satisfying the requisite exogeneity and supporting $Wealth_{i,t-2}$ as a valid instrument for addressing endogeneity in wealth dynamics.

Counterfactual simulation model. To explore how alternative retirement–age trajectories might shape individual wealth accumulation and the broader distribution of social wealth, we take the 2020 wave of the China Family Panel Studies (CFPS)—the latest year currently available—as the baseline micro–dataset and construct an ex-ante counterfactual experiment: although the statutory reform is scheduled to start in 2025, we apply the reform path as if it started in 2021 and evaluate a stylised five-year post-implementation window (2021–2025) under 2020 socioeconomic conditions. The simulation design is summarised in Table 2.

The three scenarios differ only in the statutory retirement-age schedule. The Baseline maintains the 2020 ages—60 for men, 55 for female cadres, and 50 for female workers—and serves as the no-reform counterfactual. Scenario 1 applies a gradual schedule calibrated to the officially announced plan but re-indexed to begin in 2021 for the counterfactual window (2021–2025), thereby capturing early-window distributional effects under the assumed 2020 closure rather than realised implementation outcomes. Scenario 2 imposes an immediate jump in 2021 to the long-run targets (63/58/55) and is used as a terminal, stress-test contrast to anchor an upper bound for short-run distributional impacts within the same five-year window. Thus, all three scenarios are evaluated over an identical five-year horizon, and the contrasts are interpreted as policy-path differences over a common time frame rather than long-horizon forecasts.

Table 2 Specification of delayed-retirement policy scenarios, 2021–2025.

Scenario	Brief label	Policy mechanism and assumptions (2021–2025)
Baseline	Status quo	Statutory retirement ages remain at their 2020 levels—60 years for men, 55 years for female cadres and 50 years for female workers. This trajectory serves as the reference benchmark for all counter-factual comparisons.
Scenario 1	Phased reform	In line with the official *Schedule of Incremental Increases in the Statutory Retirement Age* (MoHRSS (China) 2024), the retirement-age path is calibrated to the official month-by-month schedule but re-indexed to begin in 2021 for the counterfactual window; in implementation, the increment profile is mapped into the annual statutory ages used in our simulation steps (2021–2025).
Scenario 2	Shock reform	To stress-test a radical alternative, we assume that in 2021 the statutory retirement age jumps immediately to the long-term policy targets and then remains fixed: 63 years for men, 58 years for female cadres and 55 years for female workers.

All scenarios are evaluated over a counterfactual five-year window (2021–2025) using the 2020 CFPS wave as the baseline dataset; although the statutory reform is scheduled to start in 2025, we re-index the reform path to begin in 2021 to conduct an ex-ante evaluation under 2020 socioeconomic conditions.

For like-for-like comparisons across scenarios, the empirical model, estimation sample, and variable definitions are held fixed; differences therefore reflect only the retirement-age paths specified in Table 2. We deliberately define the scenario horizon as 2021–2025 for all scenarios; extending the horizon to 2040 would require strong assumptions about macroeconomic and demographic dynamics and complete contribution histories that the available microdata cannot support.

Based on individual status in 2020, this study employs the Mesa framework to construct an individual-level agent-based model that simulates changes in individual wealth accumulation under different scenarios, thereby revealing the impact of delayed retirement policies on social wealth distribution. Mesa is particularly suited to this task because it supports heterogeneous agent dynamics and allows sequential updates of employment status, income source, and individual wealth in each period. This design makes it transparent how a policy-induced change in the statutory retirement age alters the timing of the wage-to-pension switch at the micro level and how these effects aggregate into the distribution of social wealth. Moreover, Mesa is open-source, with full source code and documentation hosted on GitHub (Project Mesa Developers 2020) which ensures transparency, reproducibility, and extensibility for future research.

1. Initial state setup. The individual wealth status from 2020 is used as the initial condition for the agent–based model. The retirement schedule in the initial state is adjusted based on each scenario. For instance, in scenario 1, the official month-by-month increment profile is mapped into annual statutory ages consistent with our annual simulation step, whereas in scenario 2, it is immediately postponed by five years.

Using the Mesa framework, an 'IndividualAgent' class is defined, inheriting from 'mesa.Agent', to represent each individual. Each agent has the following attributes: current wealth level, planned retirement year, current wage level, pension income, personal consumption level, total household assets, household income, self–assessed health status, urban–rural status, gender and regional attributes. The agent class also defines a step method, which executes the behavior logic at each time step, including employment status assessment, income adjustment and wealth updates.

2. Dynamic wealth accumulation simulation. Based on the individual wealth accumulation model, the simulation tracks individual wealth accumulation from 2021 to 2025. The specific steps are as follows:

i. Employment–Status Determination: According to the individual's age and the scenario settings, it is determined whether the retirement condition is met. If the individual has not reached retirement age, they retain their current employment status; if they have reached retirement age, their wage income is replaced by pension income.
ii. Income adjustment: For individuals meeting the retirement condition, their pension level is estimated using a conditional–distribution kernel–density function, with wage income replaced by the corresponding pension income.
iii. Wealth update: The estimated dynamic panel data model is used to derive the individual's wealth level for the next period.
iv. Iterative update: The simulated wealth level is set as the new initial state and the steps above are repeated until the simulation for 2025 is completed. The Scheduler function in Mesa allows for the definition of agent execution order and time steps, facilitating dynamic simulation.
v. Short-run behavioural closure. Over the common 2021–2025 horizon, we hold each agent's occupation and CPI-deflated wage at its 2020 level until the retirement event is triggered by the scenario-specific statutory age. This constant wage/occupation closure isolates policy-path timing effects from wage bargaining, job mobility, or general-equilibrium feedbacks that cannot be credibly identified over such a short window with the available microdata.
vi. Compliance assumption. We implement a policy-compliance rule whereby all individuals follow the statutory schedule implied by each scenario; voluntary non-adoption is not modelled at this early roll-out stage. Scenarios 1–2 should therefore be read as upper-bound contrasts relative to partial compliance.

3. Result saving and analysis. Data are collected using the 'DataCollector' and saved for each simulation period. The changes in individual wealth accumulation are aggregated, enabling the analysis of variations in social wealth distribution across different scenarios. By comparing the simulation results of the baseline scenario with those of the two delayed retirement policy scenarios, the mechanism through which delayed retirement policies influence social wealth distribution via individual wealth accumulation is identified.

By constructing an agent–based model within the Mesa framework (as shown in Fig. 1), individual wealth accumulation trajectories are systematically simulated under different delayed retirement schemes, assessing the potential impact of delayed retirement policies on social wealth distribution. This approach not only complements empirical analyses based on historical data but also provides policymakers with scientific projections, helping them understand the possible economic effects of delayed retirement policies under different implementation paths. An in-sample fit check comparing the simulated and observed 2020

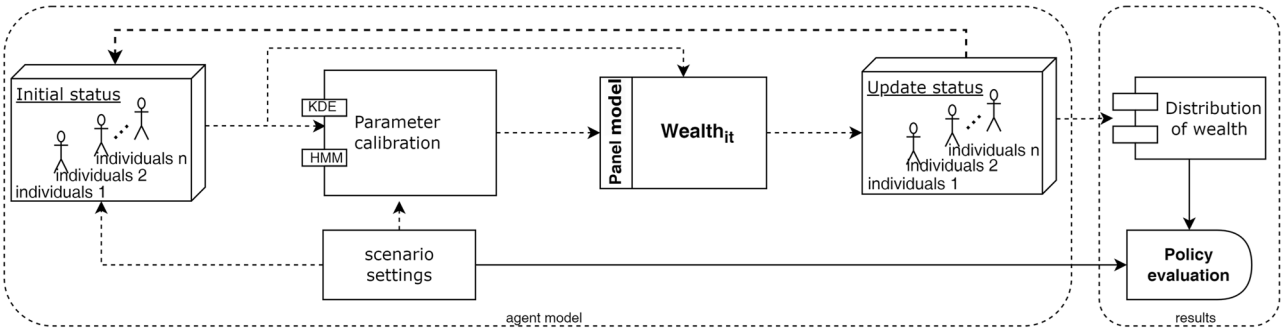


Fig. 1 Analytical framework for evaluating delayed retirement and wealth distribution.

personal wealth distributions is reported in Supplementary Fig. S1.

Dynamic Parameter Calibration. To estimate the conditional expected pension level given wage income and occupation category ($d \in \{1, \ldots, 8\}$), this study uses a mixed–kernel density estimator that combines a Gaussian kernel for continuous variables with the Aitchison–Aitken kernel for unordered categorical variables (Li and Racine 2007). Within China's pension architecture, wages determine the contribution base of the Urban Employees' Basic Pension, while occupational classification captures sector-specific plan rules and replacement–rate differentials (Wang et al. 2023). Recent empirical evidence reports a wage–pension elasticity of approximately 0.35–0.48 for higher earners (Li et al. 2020) and a public–sector premium of roughly 25 per cent relative to the non–public sector (Wang and Huang 2023). These findings substantiate the choice of wage and occupation as conditioning variables in the mixed–kernel framework.

1. Kernel density estimation of expected pension values. Under the Chinese pension system, wages determine the contribution base of the Urban Employees' Basic Pension, while occupation categories capture scheme–specific replacement–rate differentials (Wang et al. 2023). Empirical evidence shows that (i) the elasticity of pensions with respect to wages for high–income groups is about 0.35–0.48 (Li et al. 2020); and (ii) pensions in the "public sector" enjoy a ~ 25% premium over the "non–public sector" (Wang and Huang 2023).

To estimate $E[p|w,d]$ for a given wage ww and occupation $d \in \{1, \ldots, 8\}$, we employ a Gaussian–Aitchison–Aitken mixed kernel estimator (Aitchison and Aitken 1976) that accommodates one continuous and one discrete dimension.

For the sample $n\{(p_i, w_i, d_i)\}_{i=1}^{n}$––with p_i pensions, w_i CPI–deflated wages and d_i occupation codes––the joint density is

$$\hat{f}(p,w,d) = \frac{1}{nh_p h_w} \sum_{i=1}^{n} K\left(\frac{p-p_i}{h_p}\right) K\left(\frac{w-w_i}{h_w}\right) L(d,d_i;\lambda)$$

where $K(\cdot)$ is Gaussian and h_p, h_w are bandwidths; the discrete kernel follows Aitchison and Aitken (1976):

$$L(d,d_i;\lambda) = \begin{cases} 1-\lambda, d = d_i \\ \frac{\lambda}{m-1}, d \neq d_i, \end{cases} \quad 0 \le \lambda \le 1, m = 8$$

which satisfies $\sum_{d_i} L(d,d_i;\lambda) = 1$.

The estimated conditional density of pensions, given (w,d), is

$$\hat{f}_{P|W,D}(p|w,d) = \frac{\hat{f}(p,w,d)}{\int_{-\infty}^{\infty} \hat{f}(u,w,d)du}$$

and the corresponding conditional expectation is

$$\hat{E}[p|w,d] = \int_{-\infty}^{\infty} p\hat{f}_{P|W,D}(p|w,d)dp = \frac{\sum_{i=1}^{n} p_i K\left(\frac{w-w_i}{h_w}\right) L(d,d_i;\lambda)}{\sum_{i=1}^{n} K\left(\frac{w-w_i}{h_w}\right) L(d,d_i;\lambda)}$$

Smoothing parameters are selected via (Silverman 2018) two–step cross–validation: leave–one–out least–squares CV for (h_p, h_w), followed by log–likelihood CV for λ. The optimum is ($p^* = 3.812$kCNY, $h_w^* = 5.106$kCNY, $\lambda^* = 0.221$).

In the short-run closure adopted here (2021–2025), pre-retirement wage and occupation are treated as sufficient statistics for pension determination. This reflects the earnings-related and scheme-differentiated design of China's pension system, where benefit levels are closely tied to observed wages and occupational categories. Accordingly, the conditional expectation recovered by the mixed-kernel estimator is used as the pension value when retirement is triggered in the simulation.

In the simulation, wages are mapped to pensions as follows: the mixed-kernel estimator recovers the conditional expectation of first-period pension receipts given pre-retirement wage and occupation (CFPS 2020). In the simulation, when an agent retires, wage income is replaced by this expected pension amount. We do not endogenise additional contribution months/years arising from delayed retirement within 2021–2025; this constitutes a deliberate short-run, partial-equilibrium closure adopted to preserve like-for-like comparability across scenarios, with related limitations discussed in §6.

Varying each parameter by ±20% changes the mean absolute deviation of the expected–pension surface by ≤ 3%, confirming robustness (Supplementary Table S2). The mixed–kernel estimator is implemented on the 2020 wave of the CFPS dataset and the resulting wage–by–occupation expectation surface is visualized in Fig. 2 as a heat–map with colour shading and contour lines.

2. Hidden Markov model and viterbi algorithm for health state transitions. To predict changes in individual health status, this study uses health state transition data from 2010 to 2020 and applies a Hidden Markov Model (HMM) to estimate transition probabilities (see Fig. 3). The Viterbi algorithm is then used to identify the most likely sequence of health states for the next period.

i. Definition of hidden Markov model. A hidden Markov model (HMM) is typically represented by the triplet $\lambda = (A, B, \pi)$ and is defined as follows:

Hidden State Set: $S = \{S_1, S_2, \ldots, S_N\}$, representing an individual's health states.

Observation State Set: $O = \{O_1, O_2, \ldots, O_T\}$, where each state in the sequence represents an observation from the set.

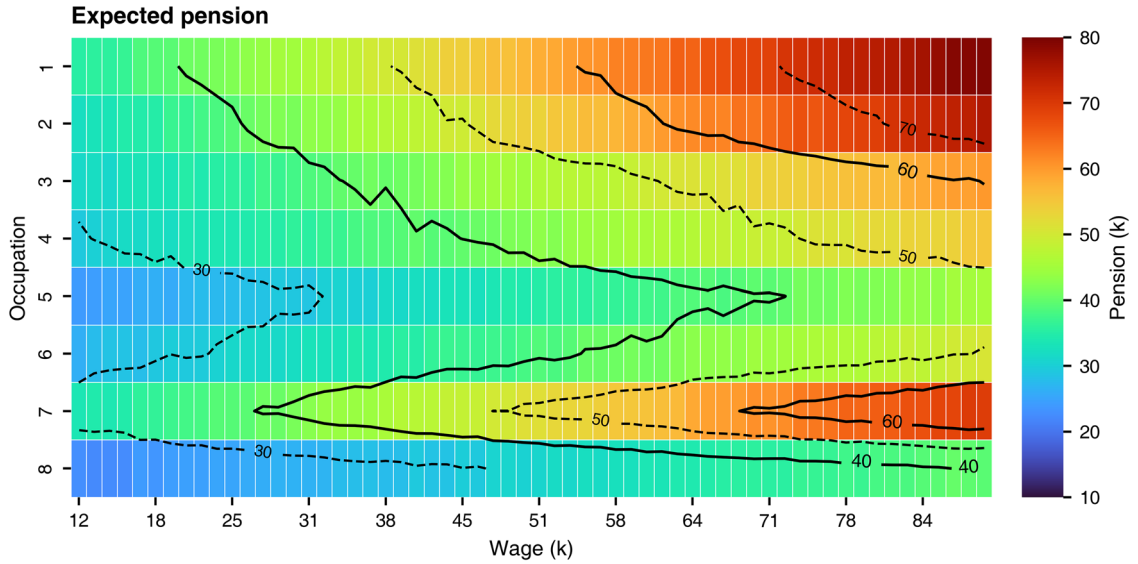


Fig. 2 Conditional expected pension by wage and occupation (mixed-kernel estimate). Note: i The color blocks indicate the expected pension (in 1000 yuan) for each occupation and wage group. Black contour lines represent isolines of equal pension (in 1000 yuan); dashed lines show minor intervals, while solid lines indicate major thresholds. ii Occupation codes 1–8 are defined as follows: 1 = Heads of government agencies, CPC organizations, enterprises, or public institutions; 2 = Professional and technical personnel; 3 = Clerical and related personnel; 4 = Commercial and service workers; 5 = Agricultural, forestry, animal husbandry, fishery and water conservancy production personnel; 6 = Production and transport equipment operators and related personnel; 7 = Military personnel; 8 = Other occupations not elsewhere classified.

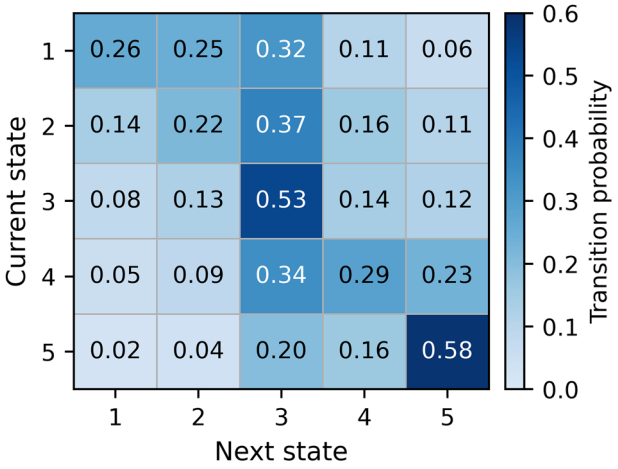


Fig. 3 Health state transition probability matrix. Note: Values denote one-period transition probabilities between discrete health states and are rounded to two decimal places for visual clarity. The underlying transition matrix is identical to that used in the analysis.

State Transition Matrix: $A = \{a_{ij}\}$, defining the probability of transitioning from state S_i *to* S_j, i.e.,

$$a_{ij} = P\left(S_{t+1} = S_j, |, S_t = S_i\right)$$

Observation Probability Matrix: $B = \{b_j(k)\}$, defining the probability of observing state O_k given the hidden state S_j, i.e.,

$$b_j(k) = P\left(O_t = O_k, |, S_t = S_j\right)$$

Initial State Distribution: $\pi = \{\pi_i\}$, defining the probability of being in state S_i at the initial time $t = 1$, i.e.,

$$\pi_i = P\left(S_1 = S_i\right)$$

ii Viterbi algorithm derivation. Given an observation sequence $O = \{O_1, O_2, \ldots, O_T\}$, the Viterbi algorithm finds the most likely hidden state sequence $S = \{S_1, S_2, \ldots, S_T\}$, maximizing the conditional probability $P(S, |, O, \lambda)$. The Viterbi algorithm, a dynamic programming approach, is widely used for sequence prediction (Rabiner 1989). The derivation includes the following steps:

a. Initialization: For each hidden state S_j, calculate the optimal path probability for the first period $\delta_1(j)$:
$$\delta_1(j) = \pi_j \cdot b_j(O_1)$$
where π_j is the initial state probabilityand $b_j(O_1)$ is the probability of observing O_1 in state S_j.
b. Recursion: For each time $t = 2, 3, \ldots, T$ and each state S_j, calculate the maximum path probability reaching state S_j:
$$\delta_t(j) = \max_{i=1,\ldots,N}\left(\delta_{t-1}(i) \cdot a_{ij}\right) \cdot b_j(O_t)$$
and record the source state of the best forward path $\psi_t(j)$:
$$\psi_t(j) = \max_{i=1,\ldots,N}\left(\delta_{t-1}(i) \cdot a_{ij}\right)$$
c. Termination: At the final time point $t = T$, find the maximum path probability P^* of the final state S_T:
$$P^* = \max_{j=1,\ldots,N} \delta_T(j)$$
and determine the state with the maximum probability:
$$S_T^* = \max_{j=1,\ldots,N} \delta_T(j)$$
Starting from(S_T^*), backtrack through the recorded path $\psi_t(j)$ to obtain the optimal state sequence from $t = 1$)*to*$(t = T$.
d. Health State Prediction: Using individual health state observations $O = \{O_1, O_2, \ldots, O_6\}$, the Viterbi algorithm calculates the optimal state path $S = \{S_1, S_2, \ldots, S_6\}$. Based

on the state transition probabilities a_{ij}, the most likely next–period state S_7 can be inferred from the last observed stateS_6:

$$S_7 = \max_{j=1,\ldots,N}\left(\delta_6(i)\cdot a_{ij}\right)$$

where $\delta_6(i)$ is the maximum path probability of reaching state S_i in period 6.

Empirical Results and Analysis

Individual wealth accumulation. The drop in the lagged–wealth coefficient from 0.482 (OLS) to 0.322 (2SLS) does not necessarily contradict classic Nickell bias. Nickell (1981) shows that, in a short–T fixed–effects (FE) dynamic panel, the within–group OLS estimator is typically downward–biased because the demeaned lagged variable is mechanically negatively correlated with the demeaned error term. Yet later studies (Hayakawa 2009) demonstrate that the direction of bias can flip if the idiosyncratic shock is positively serially correlated or if time–varying unobservables are imperfectly purged by the individual dummy. In our setting, household wealth is known to exhibit highly persistent shocks (e.g., capital–gain surprises) that raise both current errors and lagged wealth, creating a positive correlation and hence an upward FE–OLS bias. Instrumenting the lag with $Wealth_{i,t-2}$ severs this link, so the coefficient—now 0.322—falls to a more conservative but still significant estimate, consistent with Anderson and Hsiao (1981) prescription for eliminating dynamic bias. These estimates are summarized in Table 3.

By contrast, the coefficients on income (from 0.178 to 0.318) and consumption (from 0.207 to 0.292) increase after IV estimation. The upward revision is consistent with classical attenuation from measurement error and simultaneity: noisy self–reported earnings/expenses and contemporaneous wealth–management decisions both drive OLS toward zero. Valid external instruments (*RetAge_Hit*, *PeerWage*) break those correlations, revealing stronger true effects—an outcome widely documented in the household–finance literature (Campbell and Taksler 2003; Fagereng et al. 2020).

Urban residency, male gender and higher age each display a statistically significant positive association with household wealth, whereas poorer self–rated health shows a significant negative association; these patterns remain robust under both FE–OLS and FE–2SLS estimators, confirming that the demographic controls are functioning as intended in the model.

In sum, after applying the FE–2SLS estimator to address potential measurement error and other endogeneity, we obtain economically larger and more credible coefficients on income and consumption, reflecting a more pronounced true effect, while the coefficient on lagged wealth is reduced to a more realistic magnitude.

The three instruments—statutory–retirement eligibility (*RetAge_Hit*), peer–average wage (*PeerWage*) and the second lag of wealth ($Wealth_{i,t-2}$)—all satisfy the relevance criterion.

i. In the first–stage regressions, *RetAge_Hit* lowers income by −5.175 (SE 0.850) and consumption by −0.841 (SE 0.265), whereas *PeerWage* raises them by 0.752 (SE 0.140) and 0.128 (SE 0.038), respectively.
ii. The corresponding excluded–instrument F–statistics are 85.7 (income) and 28.9 (consumption). For lagged wealth, $Wealth_{i,t-2}$ enters with a coefficient of 0.457 (0.035) and an F–statistic of 76.4.
iii. All Kleibergen–Paap rk Wald F values (43.2, 22.1, 45.8) exceed the Stock–Yogo 10% critical value of 16.4, ruling out weak–instrument concerns (Stock and Yogo 2005).

Table 3 Fixed-Effects OLS, FE-2SLS and Instrument Diagnostics.

	(1) FE-OLS	(2) FE-2SLS
Panel a: Second-stage coefficients		
$Wealth_{i,t-1}$	0.482 (0.042) ***	0.322 (0.056) ***
Income	0.178 (0.049) ***	0.318 (0.108) ***
Consumption	0.207 (0.058) ***	0.292 (0.087) ***
Urban	5.108 (2.505) **	5.251 (3.142) *
Male	2.812 (1.119) **	2.956 (1.280) **
Age	0.260 (0.122) **	0.244 (0.126) *
Self-rated health	−1.368 (0.551) **	−1.798 (0.608) ***
Constant	8.312 (5.561)	9.218 (5.211) *
Within R^2	0.694	0.678
Observations	65 436	65 436
Individuals	10 906	10 906
Individual FE	Yes	Yes
Region-Year FE	Yes	Yes
Panel b. First-stage diagnostics		
Endogenous var. = Income		
Coefficient on *RetAge_Hit*	—	-5.175 (0.850) ***
Coefficient on *PeerWage*	—	0.752 (0.140) ***
Joint F-stat, excluded IVs	—	85.712
Kleibergen-Paap rk Wald F	—	43.2 > 16.4
Endogenous var. = Consumption		
RetAge_Hit	—	-0.841 (0.265) ***
PeerWage	—	0.128 (0.038) ***
Joint F-stat, excluded IVs	—	28.933
Kleibergen-Paap rk Wald F	—	22.1 > 16.4
Endogenous var = $Wealth_{t-1}$		
Coefficient on $Wealth_{t-2}$	—	0.457 (0.035) ***
First-stage F (single IV)	—	76.404
Kleibergen-Paap rk Wald F	—	45.8 > 16.4
Panel c: Instrument-validity tests		
Wooldridge AR(1), (panel)	F = 1.202, p = 0.310	F = 1.210, p = 0.302
Arellano-Bond AR(2), first-diff.		P = 0.456

Robust standard errors (in parentheses) are clustered at the individual level; ***, **, * denote significance at the 1%, 5% and 10% levels.
All regressions include individual and region-year fixed effects.
Instruments: *RetAge_Hit*, *PeerWage* and $Wealth_{i,t-2}$. Kleibergen-Paap rk Wald F-statistics for each endogenous variable exceed the Stock-Yogo 10% critical value (16.4), indicating strong instruments.
The Arellano-Bond AR(2) test is applied to the first-differenced residuals of the dynamic first-stage regression. The null hypothesis is "no second-order serial correlation." A *p*-value above conventional significance levels confirms that using $Wealth_{t-2}$ as an internal instrument is valid.
The model uses three excluded instruments for three endogenous regressors; because the number of instruments equals the number of endogenous variables, the system is exactly identified and over-identification tests (e.g., Hansen J) are not applicable.

To bolster the exclusion restriction we rely on three safeguards.

i. High–dimensional controls. Region–by–year fixed effects and flexible age polynomials absorb any macro or cohort shocks that could let either *RetAge_Hit* or *PeerWage* affect wealth directly (e.g., local housing booms or age–specific policy reforms). Residual variation in the instruments therefore works primarily through income and consumption.
ii. Economic mechanism and prior evidence. Statutory retirement age acts as an exogenous labour–supply shock; in systems that disburse pensions monthly, it influences wealth only via earnings and spending (Coe and Zamarro 2011; Etgeton et al. 2023). Likewise, peer wages reflect village–level labour–market conditions that shift an individual's wage offer but, conditional on region–year dummies, have no independent pathway to household portfolios (Stroebel and Vavra 2019).

Table 4 Robustness Checks.

Variable	2SLS					LIML	GMM	IPW
	Full	Urban	Rural	Male	Female			
$Wealth_{i,t-1}$	0.322	0.305	0.445	0.295	0.315	0.330	0.380	0.290
	(0.056)***	(0.138)**	(0.092)***	(0.118)**	(0.082)***	(0.069)***	(0.082)***	(0.050)***
Income	0.318	0.320	0.390	0.325	0.315 (0.211)	0.355	0.290	0.340
	(0.108)***	(0.125)**	(0.120)**	(0.145)**		(0.110)***	(0.139)**	(0.130)**
Consumption	0.292	0.270	0.410	0.278	0.310	0.260	0.210 (0.145)	0.315
	(0.087)***	(0.158)*	(0.305)	(0.102)***	(0.140)**	(0.140)*		(0.090)***
Joint F (Income eq.)	85.712	41.269	36.585	38.474	33.725	—	—	—
Joint F (Consumption eq.)	28.933	18.636	19.164	17.220	16.908	—	—	—
observations	65436	29446	35990	30755	34681	65436	65436	65436
Individuals	10906	4908	5998	5126	5780	10906	10906	10906

Joint F (Income eq.) and Joint F (Consumption eq.) are Wald *F*-statistics for the two excluded instruments (*RetAge_Hit* and *PeerWage*) in the first-stage regressions for income and consumption, respectively. All values exceed the Stock–Yogo 10% critical value of 16.4, indicating no weak-instrument concerns across subsamples.
Because the model is exactly identified (three excluded instruments for three endogenous regressors), point estimates under LIML and two-step GMM coincide with those from 2SLS; they are reported to demonstrate robustness to small-sample bias (LIML) and to heteroskedasticity (GMM).
Stabilised IP weights are obtained from a 2010 logit of sample retention using gender, hukou status, age, schooling, marital status, baseline net assets, income and self-rated health; weights are trimmed at the 1st–99th per centiles (mean≈1.02, SD ≈ 0.19).
***, **, * denote significance at the 1%, 5%, 10% levels, respectively.

iii. Serial–correlation diagnostics for the internal instrument. The Wooldridge AR(1) test yields $p \approx 0.30$ and by construction the use of $Wealth_{i,t-2}$ to instrument $Wealth_{i,t-1}$ satisfies the Anderson–Hsiao requirement of no AR(2) error correlation (Anderson and Hsiao 1981).

Because the system is exactly identified, formal over–identification tests are infeasible; nevertheless, the combination of strong first–stage statistics, rich fixed effects and well–documented policy shocks provides persuasive evidence that *RetAge_Hit*, *PeerWage* and $Wealth_{i,t-2}$ are all powerful and exogenous, validating the FE–2SLS estimates.

Table 4 reports a battery of robustness checks that confirm the stability of our instrumental–variable (IV) results.

First, the 2SLS estimates are re–estimated on key subsamples. Coefficients on lagged wealth, current income and current consumption remain positive and (with two minor exceptions) statistically significant at the 5% level or better across urban, rural, male and female groups. The wealth–persistence effect is markedly stronger in rural areas than in cities (lagged–wealth coefficient 0.445 versus 0.305), consistent with tighter credit constraints and heavier reliance on self–insurance in rural China (Giles and Mu 2007). This pattern indicates that our main findings are not driven by any single demographic segment.

Second, we probe estimator choice. Because the model is exactly identified (three excluded instruments for three endogenous regressors), point estimates from limited–information maximum likelihood (LIML) and two–step heteroskedasticity–robust GMM coincide almost exactly with 2SLS (e.g., lagged–wealth coefficients: 0.330 under LIML, 0.380 under GMM and 0.322 under 2SLS). Coupled with joint first–stage F–statistics well above the Stock–Yogo 10% critical value (85.7 for the income equation, 28.9 for the consumption equation), these results rule out weak–instrument concerns (Murray 2006).

Third, potential attrition bias is addressed via stabilised inverse–probability weights (IPW) derived from a 2010 logit of sample retention. The IPW–weighted 2SLS estimates are nearly unchanged (lagged–wealth coefficient 0.290), indicating that attrition conditional on observables is largely ignorable—an outcome in line with evidence from other long panels (Fitzgerald et al. 1998).

Taken together, the consistency of coefficients across subsamples, estimators and attrition adjustments strengthens confidence in our identification strategy and underscores the robustness of the estimated wealth–persistence and income–and–consumption effects.

Dynamics of wealth distribution. Between 2010 and 2020, China's wealth Gini coefficient rose from 0.591 to 0.631. Under the baseline, it is expected to reach roughly 0.664 by 2025 (Fig. 4 b). This trajectory indicates a structural linkage between the current pension regime and the evolution of wealth inequality. Existing evidence shows that pension income is highly concentrated among older households (Li et al. 2020) and that its inequality far exceeds that of labour earnings (Hanewald et al. 2021). Consistent with the cumulative–advantage mechanism, formal–sector, higher–income workers accumulate more pension wealth through longer contribution histories and higher replacement rates, whereas informal and lower–income groups obtain limited entitlements, thereby widening later–life wealth gaps (Ferraro and Shippee 2009). All scenario contrasts below are reported over the same five-year window (2021–2025) and should be interpreted as policy-path differences within a common period.

Under the gradual retirement-age schedule calibrated to the official plan but re-indexed to start in 2021 (scenario 1), the Gini

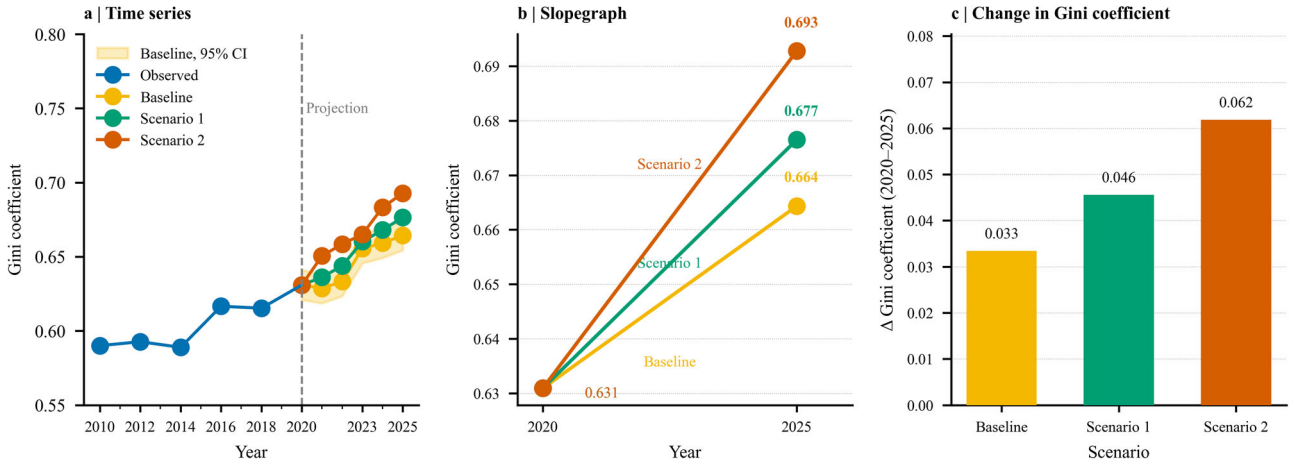


Fig. 4 Multi-scenario projection of the wealth Gini coefficient under different pension reform paths, 2010–2025. Note: **a** shows the observed Gini coefficients for 2010–2020 and projected values for 2021–2025 under the baseline, Scenario 1 and Scenario 2; the vertical dashed line marks the start of the projection period, and the shaded band indicates the 95% confidence interval. **b** compares the 2020 and 2025 Gini coefficients for each scenario using a slope graph. **c** summarizes the change in the Gini coefficient from 2020 to 2025 under each scenario. The lines, markers and bars correspond to Observed, Baseline, Scenario 1 and Scenario 2 as indicated in the figure legend. Common counterfactual horizon under 2020 conditions: 2021–2025.

Table 5 Urban–health–gender hierarchical Gini decomposition under three delayed retirement scenarios, 2021–2025 (baseline year: 2020).

Total Gini		Between Urban	Between Health within Urban	Between Gender within U-H	Bottom pure Within (U-H-G)
Panel a: Baseline					
2020	0.631	0.243	0.103	0.047	0.607
2021	0.629	0.243	0.101	0.050	0.606
2022	0.633	0.245	0.099	0.047	0.610
2023	0.656	0.246	0.093	0.051	0.610
2024	0.659	0.254	0.096	0.046	0.605
2025	0.664	0.252	0.096	0.058	0.594
Panel b: Scenario1					
2020	0.631	0.243	0.103	0.047	0.607
2021	0.636	0.262	0.101	0.058	0.579
2022	0.644	0.284	0.106	0.058	0.552
2023	0.660	0.311	0.113	0.068	0.508
2024	0.668	0.327	0.121	0.066	0.486
2025	0.677	0.351	0.130	0.065	0.455
Panel c: Scenario2					
2020	0.631	0.243	0.103	0.047	0.607
2021	0.651	0.282	0.110	0.055	0.553
2022	0.659	0.302	0.118	0.059	0.521
2023	0.665	0.316	0.117	0.068	0.500
2024	0.683	0.339	0.125	0.072	0.464
2025	0.693	0.352	0.131	0.072	0.444

Except for the total Gini coefficient, all other values are contribution proportions.

coefficient is projected to rise from 0.631 in 2020 to about 0.677 in 2025—an increment larger than in the baseline. Thus, within our short-run, partial-equilibrium, full-compliance counterfactual, postponing retirement—absent additional redistributive adjustments in pension rules—tends to yield a higher projected wealth-inequality path than the no-delay baseline. Healthier and steadily employed workers are better placed to translate longer careers into additional earnings and pension credits, while lower–income or less–healthy individuals benefit to a much smaller extent (Chen et al. 2018), reinforcing existing stratification (Hanewald et al. 2021).

Scenario 2 models a one–off, more substantial postponement of retirement and projects the Gini coefficient to reach approximately 0.693 by 2025, the largest increase across scenarios. Compared with the baseline and the gradual scheme, scenario 2 generates a sharper short–term rise in inequality: high–income, longer–lived cohorts accumulate substantially more pension rights, whereas vulnerable groups unable to delay retirement fall further behind. This pattern suggests that, conditional on our model closure and parameterisation, larger statutory-age shifts can generate a larger short-run increase in projected wealth inequality (Ferraro and Shippee 2009). Even under a moderate delay in our counterfactual calibration, complementary reforms aimed at broader coverage and stronger redistribution may be important for mitigating the distributional risks suggested by the simulations.

The Structural Mechanisms Underlying the Evolution of Social Wealth Inequality. This study employs the stratified Gini coefficient decomposition method proposed by Mussard and Savard (2012). We conduct a multi–layer decomposition of the Gini coefficient over the counterfactual window 2021–2025 (with 2020 as the baseline year), examining the evolution of social wealth inequality under delayed retirement scenarios across three dimensions: urban–rural, health and gender. Under scenario 1 (the current gradual delayed retirement scheme), the contributions of these dimensions shift significantly. Urban–rural, health and gender differences together account for roughly 40%–60% of the overall Gini coefficient, with the remainder due to within–group inequality (the "Bottom pure within" component). This distribution of inequality partly reveals the structural mechanisms driving the evolution of social wealth inequality.

According to the simulation results (Table 5b), the delayed retirement policy induces structural changes in the Gini coefficient, leading to further stratification of social wealth. First, the widening urban–rural wealth gap is the primary driver of the rising Gini coefficient. In scenario 1, the contribution of urban–rural differences to the Gini rises from 24.3% in 2020 to 35.1% in 2025 (an increase of 10.8%). In contrast, under the baseline scenario (no delay) the urban–rural contribution fluctuates between 24.3% and 25.4% and under scenario 2 it is slightly higher than in scenario 1. Although China's urban–rural wealth gap has tended to stabilize in recent years (Li and Wan

2015), the simulation indicates that a delayed retirement policy may widen urban–rural wealth inequality to some extent. Urban workers often enjoy more comprehensive social security coverage and more formal employment positions. Delayed retirement allows this group to extend their period of wealth accumulation. In contrast, rural labourers—due to low pension security, unstable employment and a high proportion of physically demanding work—can obtain only limited gains from prolonging their working years (Giles et al. 2023).

Second, health disparities play an increasingly important role in wealth inequality. Under a no–delay scenario, health differences account for roughly 10% of the Gini coefficient for wealth inequality, whereas under the delayed retirement policy this share rises to about 13%. This suggests that delayed retirement amplifies the wealth accumulation gap between groups with different health statuses. Individuals in good health remain in the workforce longer and accumulate more wealth, whereas those in poorer health tend to exit the labour market early—missing the opportunity to prolong their working lives and accumulate additional wealth (Li et al. 2024).

Third, the impact of gender differences remains very small, and its increase is limited. Delayed retirement raises the Gini contribution of gender from around 5% to about 7%, but the overall effect is negligible, indicating that the structural impact of delayed retirement on wealth inequality between men and women is relatively mild. On the one hand, women experience more frequent career interruptions and have shorter average careers, combined with a more pronounced gender pay gap in high–income positions. These factors result in significantly lower lifetime earnings and pension contributions for women compared to men (Zhao and Zhao 2018) and consequently women's overall wealth levels are lower than men's. On the other hand, the delayed retirement policy increases women's statutory retirement age by a larger margin than men's, which helps extend women's working years and promotes greater gender wealth equality (Zhao et al. 2024). This has, to some extent, curbed the widening of gender wealth inequality. Throughout the simulation period, the gap in accumulated wealth between men and women remains largely unchanged and gender consistently accounts for only a small fraction of the Gini coefficient—a stark contrast to the pronounced changes in urban–rural and health contributions over the same period.

Overall, in our counterfactual design the statutory retirement age is mechanically postponed; under this stylised short-run closure, the simulations suggest that this timing shift can reshape the projected distribution of social wealth. The simulations indicate a stratifying pattern: projected wealth dispersion increases primarily along urban–rural and health dimensions, while the gender contribution changes comparatively little. In reality, the impact of delayed retirement policies varies across different groups. Such policies may allow already advantaged groups to benefit more, while some disadvantaged groups—due to factors such as health, occupation and livelihood—find it difficult to benefit equally and may even be further disadvantaged if they cannot extend their working lives (Berkman and Truesdale 2023).

Conclusion and Discussion

China's delayed–retirement policy has recently been announced and is scheduled to start in 2025; its macro–level socioeconomic consequences are still unfolding, and neither the policy's ultimate efficacy nor its broader social repercussions can be directly observed yet. This study examines the policy's potential impact on the distribution of social wealth and elucidates the structural dynamics that govern the evolution of wealth stratification, thereby providing empirical evidence and analytical insight for policy evaluation and the design of complementary measures within China's institutional and societal context.

This study advances the debate on delayed retirement and wealth inequality with four closely related, hierarchical findings. All findings below are derived within our ex-ante counterfactual setting (2021–2025 under 2020 baseline conditions) and a short-run, partial-equilibrium, full-compliance closure; they should be read as counterfactual policy-path differences rather than realised outcomes or long-run forecasts.

First, overall wealth inequality is projected to widen over 2021–2025 under both gradual and one-off increases in the statutory age; adjusting the pace of implementation alone may be insufficient to fully offset this pattern within our short-run closure, suggesting that complementary measures (e.g., broader coverage and incrementally adjusted replacement rates) could be considered.

Second, the projected inequality increase tends to propagate along existing stratifications: urban residents, healthier individuals and higher-income workers are more likely to convert additional working years into greater asset and pension accumulation, whereas rural, less-healthy and lower-income groups secure more limited gains.

Third, the Gini decomposition indicates that the contributions of urban–rural status and health tend to increase, while gender-related contributions remain comparatively small—suggesting that the policy may intensify regional and health-based disparities rather than create a new gender divide.

Fourth, we embed a micro-estimated wealth-accumulation equation in an agent-based counterfactual simulation to project distributional outcomes, illustrating how the reform could reinforce cumulative advantage in late-life wealth; this approach offers a transparent ex-ante framework for exploring distributional risks.

The study contributes an additional vantage point for both theoretical inquiry and policy analysis. First, our counterfactual simulations suggest that delayed retirement can amplify wealth differentiation through existing structural factors—most notably urban–rural residence and health status—within the Chinese context. Second, by embedding micro-behavioural parameters in a counterfactual agent-based simulation, the research develops a scalable ex-ante evaluation framework that can be used to explore potential policy effects under explicit assumptions, before extensive longitudinal data become available. On the policy front, the simulations indicate that delayed retirement may widen overall wealth inequality and further differentiate outcomes across urban–rural and health strata. These findings provide suggestive, model-based evidence that may inform discussions on pension design and potential targeted supporting measures.

Several limitations merit acknowledgment. First, the agent-based simulation calibrates behavioural parameters from six CFPS waves and assumes their stability; if retirement, saving, or labour-supply behaviour shifts under different macroeconomic or labour-market regimes, the simulated distributional effects may be misestimated. Second, the counterfactual window is relatively short (2021–2025 under 2020 baseline conditions), so it cannot capture longer-run adjustments such as intergenerational wealth transfers or broader general-equilibrium feedbacks that may dampen or amplify the wealth effects. Third, the analysis focuses on urban–rural residence, health status and gender, omitting other stratifiers such as educational attainment and regional economic variation, which may mask additional heterogeneity. Fourth, the model adopts a partial-equilibrium perspective and does not explicitly incorporate capital-market adjustments, firm employment strategies, or youth–labour substitution, limiting external validity. The simulation abstracts from endogenous wage

growth, occupational mobility, and benefit re-optimisation induced by the reform. Holding occupation and real wages at baseline levels and mapping pensions from observed pre-reform relationships ensure scenario comparability but may overstate effects relative to partial compliance or wage adjustments. These features are left for future work when administrative records become available.

Further research could advance along several lines. First, the analysis should extend the time horizon and embed flexible retirement behaviour, demographic evolution, capital–market adjustments and firm labour demand within an integrated general–equilibrium framework. Such an extension would illuminate the reform's long–run feedback mechanisms. Second, because our counterfactual simulations are based on the 2020 microdata baseline and are conditional on a common 2021–2025 horizon, full compliance, and fixed real wages and occupations through retirement, linking the agent-based simulation to high-frequency administrative data—such as pension account records and tax records—would enable sensitivity analyses of compliance rates and wage adjustments, alongside tighter parameter calibration and improved estimation precision. Accordingly, all results should be interpreted as counterfactual policy-path differences within a stylised five-year window (2021–2025) under 2020 baseline conditions, not as realised outcomes or long-run forecasts. Third, quantifying the interaction between delayed retirement and redistributive instruments, including progressive replacement rates or minimum–benefit guarantees, would inform the design of coherent reform packages. Fourth, applying the framework across countries or regions would test its external validity and enrich the comparative evidence base.

Data availability

The individual-level CFPS microdata analysed in this study are third-party human-subjects survey data and cannot be redistributed by the authors (including any cleaned or modified versions) under the CFPS/PKU data-use terms. Authorised researchers may obtain access to the relevant CFPS waves/modules via the official CFPS/PKU data portal. For details on publishing with CFPS data, see: https://www.isss.pku.edu.cn/cfps/en/faq/PublishwithCFPSData/index.htm.

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Author contributions

Conceptualization/Methodology: WNW, CF. Formal analysis/Visualization: XS. Investigation/Writing – review & editing: WJW. Writing – original draft: XS, WJW, WNW, CF. Approval: All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

This study is a secondary analysis of de-identified microdata from the China Family Panel Studies (CFPS). According to CFPS documentation, the CFPS project is administered under institutional ethical oversight at Peking University. The authors analysed the data under the CFPS/PKU data-use agreement and made no attempt to identify individual respondents.

Informed consent

Informed consent was obtained from all participants by the CFPS research team. The authors had no direct contact with participants and analysed de-identified secondary data; no additional consent was required for this study.

Additional information

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